
Ali Subsystem 3 - Stage 1 Feature Comparison

Power Line Noise Detection, Analysis, and Location #2

Purpose: Compare all four Dr. Talley starter recordings using the same baseline processing and feature calculations.

IMPORTANT: This script does NOT label any recording as a confirmed power-line spark. It only creates objective measurements for later detector development.

Put this file in the SAME folder as the four road_event_*.mat recordings. Then run: ali_stage1_feature_compare

```
clear;
clc;
close all;

files = dir('road_event_*.mat');
if isempty(files)
    error(['No road_event_*.mat files were found. Put this script in the ',
    ...
        'same folder as the four Dr. Talley MAT recordings.']);
end

[~, order] = sort({files.name});
files = files(order);
nFiles = numel(files);

File = strings(nFiles,1);
CenterFreq_MHz = nan(nFiles,1);
SampleRate_kSps = nan(nFiles,1);
Duration_s = nan(nFiles,1);
Gain_dB = nan(nFiles,1);
LostTotal = nan(nFiles,1);
Median_dBFS = nan(nFiles,1);
RMS_dBFS = nan(nFiles,1);
P95_dBFS = nan(nFiles,1);
P99_dBFS = nan(nFiles,1);
Peak_dBFS = nan(nFiles,1);
CrestFactor_dB = nan(nFiles,1);
FoldRange_dB = nan(nFiles,1);
StrongestPhase_deg = nan(nFiles,1);
StrongestPhase_ms = nan(nFiles,1);
SpectralPeakOverMedian_dB = nan(nFiles,1);
SpectralFlatness = nan(nFiles,1);
CandidateTransientCount = nan(nFiles,1);

fprintf('\nALI SUBSYSTEM 3 - STAGE 1 FEATURE COMPARISON\n');
fprintf('-----\n');
fprintf('These are baseline measurements, NOT confirmed spark labels.\n\n');

for k = 1:nFiles
    filename = files(k).name;
    S = load(filename);
```

```

if ~isfield(S,'eventIQ') || ~isfield(S,'fs')
    warning('%s is missing eventIQ or fs. Skipping.', filename);
    continue;
end

x = double(S.eventIQ(:));
fs = double(S.fs);
if isfield(S,'mainsFrequency')
    mainsHz = double(S.mainsFrequency);
else
    mainsHz = 60;
end

File(k) = string(filename);
CenterFreq_MHz(k) = getScalar(S,'fc',NaN)/1e6;
SampleRate_kSps(k) = fs/1e3;
Duration_s(k) = (numel(x)-1)/fs;
Gain_dB(k) = getScalar(S,'gainDB',NaN);
LostTotal(k) = getScalar(S,'lostTotal',NaN);

% 1) Baseline preprocessing
x = x - mean(x);
cutoffHz = min(50e3, 0.45*fs);
try
    xFilt = lowpass(x, cutoffHz, fs);
catch
    warning('lowpass() failed for %s. Using unfiltered IQ.', filename);
    xFilt = x;
end

env = abs(xFilt);
level_dBFS = 20*log10(env + eps);

% 2) Time-domain features
Median_dBFS(k) = median(level_dBFS);
RMS_dBFS(k) = 20*log10(sqrt(mean(abs(xFilt).^2)) + eps);
P95_dBFS(k) = simplePercentile(level_dBFS,95);
P99_dBFS(k) = simplePercentile(level_dBFS,99);
Peak_dBFS(k) = max(level_dBFS);
rmsEnv = sqrt(mean(env.^2));
CrestFactor_dB(k) = 20*log10((max(env)+eps)/(rmsEnv+eps));

% 3) Nominal 60 Hz cycle folding
samplesPerCycle = round(fs/mainsHz);
nCycles = floor(numel(xFilt)/samplesPerCycle);
if nCycles >= 2
    xUse = xFilt(1:nCycles*samplesPerCycle);
    powerMatrix = reshape(abs(xUse).^2, samplesPerCycle, nCycles);
    avgCyclePower = mean(powerMatrix,2);
    avgCycle_dB = 10*log10(avgCyclePower + eps);
    FoldRange_dB(k) = max(avgCycle_dB) - min(avgCycle_dB);
    [~, idxMax] = max(avgCycle_dB);
    StrongestPhase_ms(k) = (idxMax-1)/fs*1000;
    StrongestPhase_deg(k) = mod((idxMax-1)/samplesPerCycle*360,360);
end

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end

% 4) Frequency-domain features
nfft = min(8192, max(1024, 2^nextpow2(min(numel(xFilt), 8192))));
windowLength = min(4096, numel(xFilt));
if windowLength >= 256
    overlap = floor(0.5*windowLength);
    [Pxx,~] = pwelch(xFilt, hann(windowLength, 'periodic'), overlap,
nfft, fs, 'centered');
    Pxx = real(Pxx(:));
    Pxx(Pxx <= 0) = eps;
    Pxx_dB = 10*log10(Pxx);
    SpectralPeakOverMedian_dB(k) = max(Pxx_dB) - median(Pxx_dB);
    SpectralFlatness(k) = exp(mean(log(Pxx))) / mean(Pxx);
end

% 5) Exploratory transient count
% Adaptive only. This is NOT the final detector.
medLevel = median(level_dBFS);
madLevel = median(abs(level_dBFS - medLevel));
threshold = medLevel + 6*madLevel;
above = level_dBFS > threshold;
starts = find(diff([false; above]) == 1);

if isempty(starts)
    CandidateTransientCount(k) = 0;
else
    minGap = round(0.002*fs); % merge crossings within 2 ms
    CandidateTransientCount(k) = 1 + sum(diff(starts) > minGap);
end

fprintf('%d/%d  %s\n', k, nFiles, filename);
fprintf('  Median level:           %8.2f dBFS\n', Median_dBFS(k));
fprintf('  Peak level:               %8.2f dBFS\n', Peak_dBFS(k));
fprintf('  Crest factor:             %8.2f dB\n', CrestFactor_dB(k));
fprintf('  60 Hz fold range:         %8.2f dB\n', FoldRange_dB(k));
fprintf('  Strongest cycle phase:     %8.2f deg (%6.3f ms)\n',
StrongestPhase_deg(k), StrongestPhase_ms(k));
fprintf('  Spectral peak/median:      %8.2f dB\n',
SpectralPeakOverMedian_dB(k));
fprintf('  Spectral flatness:         %8.4f\n', SpectralFlatness(k));
fprintf('  Candidate transients:      %8d\n\n',
CandidateTransientCount(k));
end

T = table(File, CenterFreq_MHz, SampleRate_kSps, Duration_s, Gain_dB,
LostTotal, ...
    Median_dBFS, RMS_dBFS, P95_dBFS, P99_dBFS, Peak_dBFS,
CrestFactor_dB, ...
    FoldRange_dB, StrongestPhase_deg, StrongestPhase_ms, ...
    SpectralPeakOverMedian_dB, SpectralFlatness,
CandidateTransientCount);

disp(T);

```

```

outFile = 'ali_stage1_feature_table.csv';
writetable(T,outFile);
fprintf('Saved feature table to: %s\n', outFile);

figure('Name','Ali Stage 1 - Recording Level Comparison');
bar(categorical(shortNames(File)), Median_dBFS);
ylabel('Median received level (dBFS)');
title('Median Received Level by Recording');
grid on;

figure('Name','Ali Stage 1 - 60 Hz Fold Comparison');
bar(categorical(shortNames(File)), FoldRange_dB);
ylabel('Max - min average cycle level (dB)');
title('60 Hz Fold Variation by Recording');
grid on;

figure('Name','Ali Stage 1 - Spectral Concentration Comparison');
bar(categorical(shortNames(File)), SpectralPeakOverMedian_dB);
ylabel('Peak PSD - median PSD (dB)');
title('Spectral Concentration by Recording');
grid on;

fprintf('\nStage 1 complete.\n');
fprintf('Next step: interpret the table before choosing detector
features.\n');

function value = getScalar(S,fieldName,defaultValue)
    if isfield(S,fieldName)
        temp = S.(fieldName);
        if isnumeric(temp) && isscalar(temp)
            value = double(temp);
        else
            value = defaultValue;
        end
    else
        value = defaultValue;
    end
end

function p = simplePercentile(x,q)
    x = sort(x(:));
    n = numel(x);
    if n == 0
        p = NaN;
        return;
    end
    pos = 1 + (n-1)*(q/100);
    lo = floor(pos);
    hi = ceil(pos);
    if lo == hi
        p = x(lo);
    else
        p = x(lo) + (pos-lo)*(x(hi)-x(lo));
    end
end

```

```

end

function names = shortNames(fileNames)
    names = strings(size(fileNames));
    for i = 1:numel(fileNames)
        s = char(fileNames(i));
        token = regexp(s, 'road_event_\d+_(\d+)_UTC', 'tokens', 'once');
        if isempty(token)
            names(i) = string(s);
        else
            names(i) = string(token{1});
        end
    end
end
end

```

ALI SUBSYSTEM 3 - STAGE 1 FEATURE COMPARISON

 These are baseline measurements, NOT confirmed spark labels.

```

1/4 road_event_20260911_182619_107.UTC.mat
Median level:          -34.47 dBFS
Peak level:            -14.82 dBFS
Crest factor:          16.71 dB
60 Hz fold range:      3.37 dB
Strongest cycle phase: 203.49 deg ( 9.421 ms)
Spectral peak/median:   54.20 dB
Spectral flatness:      0.0002
Candidate transients:   0

2/4 road_event_20260911_184400_330.UTC.mat
Median level:          -51.06 dBFS
Peak level:            -33.13 dBFS
Crest factor:          16.74 dB
60 Hz fold range:      1.34 dB
Strongest cycle phase: 45.36 deg ( 2.100 ms)
Spectral peak/median:   50.00 dB
Spectral flatness:      0.0002
Candidate transients:   0

3/4 road_event_20260911_191220_112.UTC.mat
Median level:          -48.37 dBFS
Peak level:            -12.36 dBFS
Crest factor:          12.07 dB
60 Hz fold range:      0.14 dB
Strongest cycle phase: 55.08 deg ( 2.550 ms)
Spectral peak/median:   90.39 dB
Spectral flatness:      0.0000
Candidate transients:   2

4/4 road_event_20260911_210016_757.UTC.mat
Median level:          -48.63 dBFS
Peak level:            -14.76 dBFS
Crest factor:          31.59 dB

```

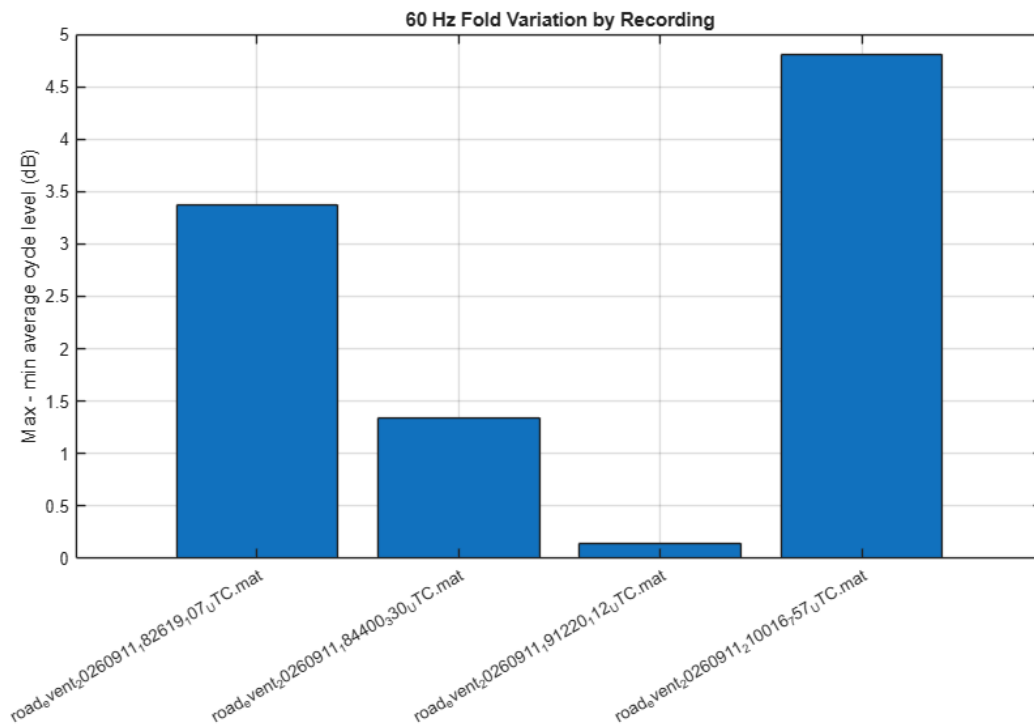
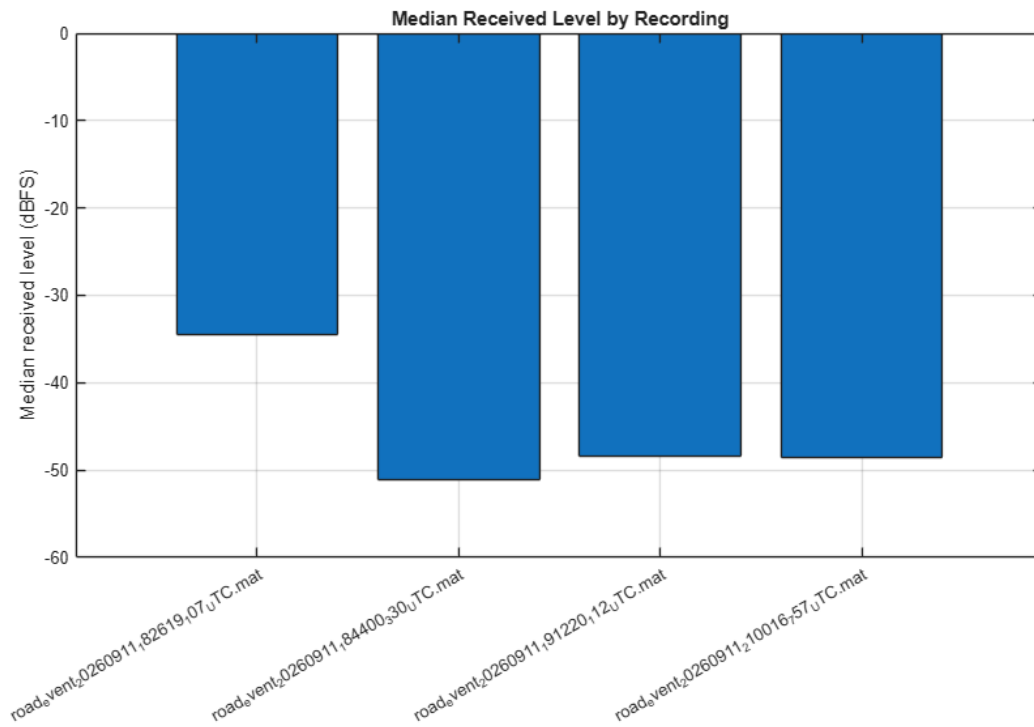
60 Hz fold range: 4.81 dB
 Strongest cycle phase: 275.85 deg (12.771 ms)
 Spectral peak/median: 64.28 dB
 Spectral flatness: 0.0001
 Candidate transients: 52

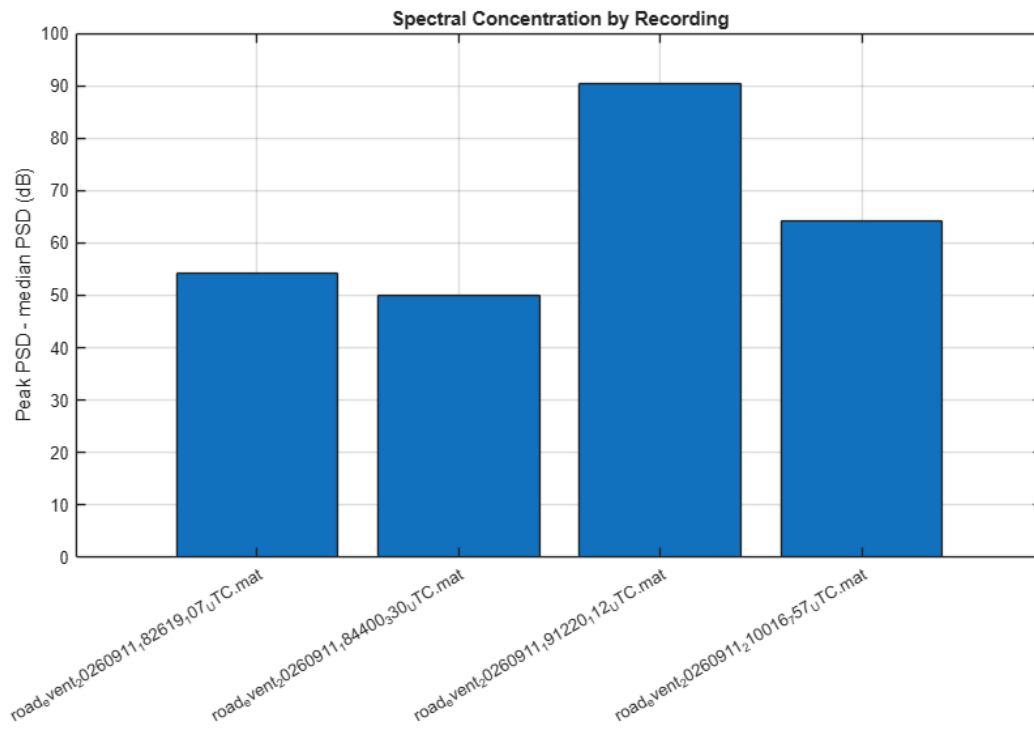
File				CenterFreq_MHz
SampleRate_kSps	Duration_s	Gain_dB	LostTotal	
Median_dBFS	RMS_dBFS	P95_dBFS	P99_dBFS	Peak_dBFS
CrestFactor_dB	FoldRange_dB	StrongestPhase_deg	StrongestPhase_ms	
SpectralPeakOverMedian_dB	SpectralFlatness	CandidateTransientCount		
"road_event_20260911_182619_107.UTC.mat"				145.9
240	12.5	28	96	-34.472
-31.528	-25.814	-23.249	-14.818	16.71
3.3735	203.49		9.4208	
54.201	0.00018637			0
"road_event_20260911_184400_330.UTC.mat"				145.9
240	10.9	28	0	-51.062
-49.876	-45.331	-43.652	-33.134	16.742
1.3397	45.36		2.1	
49.998	0.00020592			0
"road_event_20260911_191220_112.UTC.mat"				146.74
240	11.6	28	0	-48.372
-24.429	-18.109	-17.968	-12.357	12.072
0.14076	55.08		2.55	
90.391	7.8572e-07			2
"road_event_20260911_210016_757.UTC.mat"				145.9
240	15	28	0	-48.627
-46.349	-41.676	-36.733	-14.762	31.587
4.8076	275.85		12.771	
64.277	0.00011355			52

Saved feature table to: ali_stage1_feature_table.csv

Stage 1 complete.

Next step: interpret the table before choosing detector features.





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